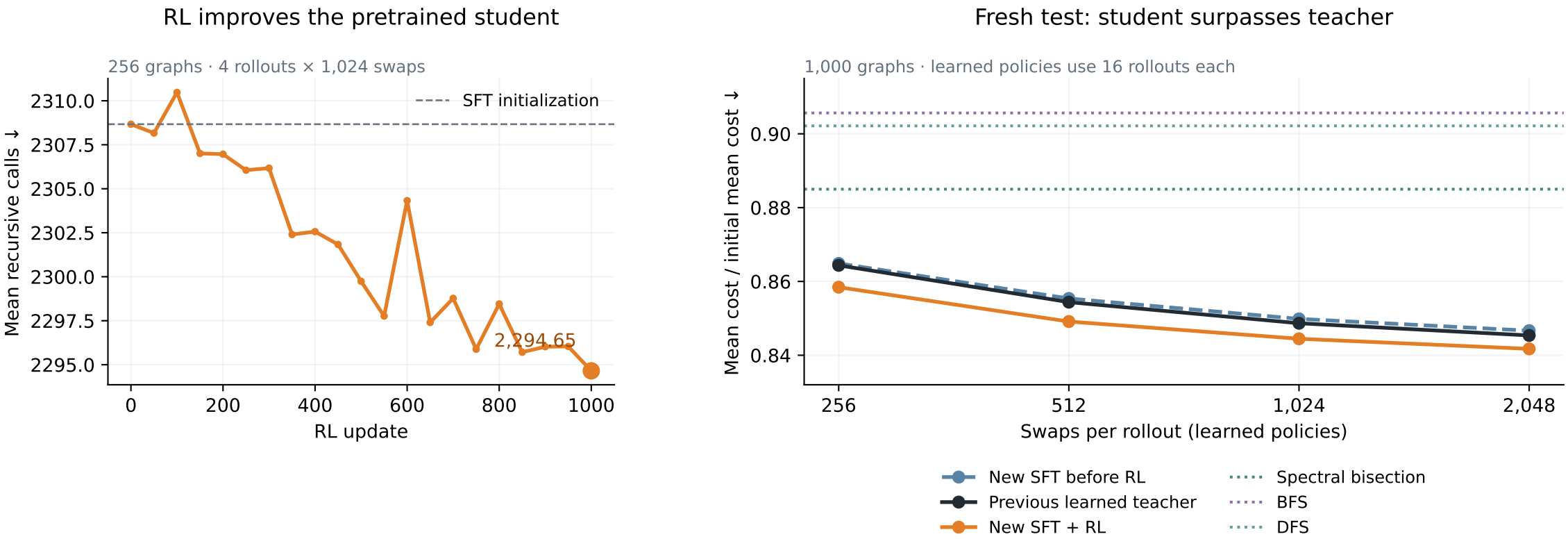
RL improves the pretrained student
256 graphs · 4 rollouts × 1,024 swaps
SFT initialization
Mean recursive calls ↓
2310.0
2307.5
2305.0
2302.5
2300.0
2297.5
2295.0
2,294.65
0
200
400
600
800
1000
RL update
Fresh test: student surpasses teacher
1,000 graphs · learned policies use 16 rollouts each
Mean cost / initial mean cost ↓
0.90
0.88
0.86
0.84
256
512
1,024
2,048
Swaps per rollout (learned policies)
New SFT before RL
Previous learned teacher
New SFT + RL
Spectral bisection
BFS
DFS
BFS, DFS and spectral bisection produce one ordering each. This compares ordering quality, not equal compute.